

Forensics CLI 101: The Aurora Case — Instructor Guide

Every challenge, answer, and accepted variant · generated 2026-08-25 from the live pack data

Quick reference

Set your own site URL and class password before handing this out. The admin handle is whatever you put in `ADMIN_HANDLES` — `instructor`, say; there is no default, and the setting ships empty.

Handles: 3–20 characters, letters/numbers/underscore/hyphen, **claimed once only**.

30 challenges · 885 points total · finds look like `FIND{ABCD2345EFGH}` and are **different for every student**.

How the difficulty fades — Act I briefs show the whole command, and the free hint repeats it. From Act II the brief names the tool but not its flags or arguments, and the exact line moves into a hint that costs XP. By the last act the brief states only the objective. A student who never buys a hint has genuinely recalled every command. Hint spend is visible in the instructor dashboard, so it reads as a difficulty signal rather than as cheating.

If a student says "it's not working"

Symptom	Cause & what to say
"My flag is rejected"	Flags are per-student — a classmate's flag never validates. Tell them to click the green flag in the terminal to copy it exactly, rather than retyping.
"Handle already claimed"	Someone took it, or they registered before and cleared their browser. Pick a new handle. (To free one, delete that player in Netlify Blobs.)
"I ran the command, nothing happened"	Check the blue path in their prompt — they may be in a different directory. Most challenges accept short or full paths, but the file must exist from where they stand. <code>cd ~</code> returns home.
"It says command not found"	If it's a real command we don't simulate (<code>top</code> , <code>nano</code>), the game says so explicitly. Genuine typos say "command not found".
"I don't know which command to use"	Intended from Act II on. Point them at the free hint first, then the Reference tab. The costed hint is the safety net, not a failure.
"The act is locked"	Each act opens after solving all but one of the previous act's challenges.
"Nothing printed"	Correct for <code>cd</code> and redirects. The game prints "(no output — in a shell, silence usually means success)".
Lost all progress	Progress is tied to the browser token. Same browser = resumes automatically. Different machine = new handle needed.

Act structure & unlocking

Act	Teaches	Challenges	Points	Unlock
Act I: First on Scene	Get your bearings on the bench: paths, prompt, and what is really in a directory	7	90	Always open
Act II: Reading the Evidence	Open the exhibits: file contents, true file types, and forensic checksums	6	120	Solve 6 of 7 in the prior act

Act III: Following the Trail	Search, find, manual pages, and the WSL bridge to the seized laptop	7	235	Solve 5 of 6 in the prior act
Act IV: The Pipeline	Pipes, redirection, filters, and multi-stage analysis	4	180	Solve 6 of 7 in the prior act
Act V: Closing the Case	Read the partition table, carry the offset, carve the container	2	180	Solve 3 of 4 in the prior act
Act VI: The Seized Laptop (Windows)	Windows CMD parity: dir /a, type, findstr, attrib, and certutil	4	80	Always open

Students may skip **one** challenge per act and still advance.

Act I: First on Scene

Get your bearings on the bench: paths, prompt, and what is really in a directory

Where Am I?

10 pts · act1-pwd · OK

Task: First rule on the bench: before you touch any evidence, put your position on the record. Print the full path of the directory you are standing in. That line is the first entry in your case log.

Answer: `pwd`

Completion: Auto-completes on: allOf

Hint 1 (free): The command is `pwd` — print working directory. Type it and press Enter.

Survey the Bench

10 pts · act1-ls · OK

Task: You cannot examine what you have not inventoried. Find out what this workstation was handed to you with: show the files and directories in the place you are standing.

Answer: `ls`

Completion: Auto-completes on: allOf

Hint 1 (free): The command is `ls` — list. On its own it lists wherever you currently are.

Also accepted: `ls .`, `ls ./`, `ls ~`, `ls /home/examiner`

Hidden in Plain Sight

15 pts · act1-hidden · OK

Task: The examiner who had this bench before you left something behind, and a plain `ls` does not show it. On Linux, a filename that starts with a dot is invisible to a normal listing. Find the hidden file in your home directory, read it, and submit the find inside.

Answer: `ls -a` then `cat .stash`

Completion: Submits a flag · flag is inside `/home/examiner/.stash`

Hint 1 (free): `ls -a` lists everything, dotfiles included.

Hint 2 (–5 XP): The file is called `.stash`. Read it with `cat .stash` and submit the recovered find.

One Level Down

15 pts · act1-cd · OK

Task: Your onboarding folder recorded a checkpoint for you. Get inside `training/level_1`, read `checkpoint_alpha.txt`, and submit the find it holds.

Answer: `cd training/level_1` then `cat checkpoint_alpha.txt`

Completion: Submits a flag · flag is inside `/home/examiner/training/level_1/checkpoint_alpha.txt`

Hint 1 (free): Use `cd training/level_1` to move down into it.

Hint 2 (–5 XP): Once inside `level_1`, run `cat checkpoint_alpha.txt` to see the find.

Sideways Move

15 pts · act1-paths · OK

Task: Now reach `training/level_2` — a SIBLING of `level_1`, not a child. If you are standing inside `level_1`, you have to go up before you can go over. Read `checkpoint_beta.txt` and submit the find.

Answer: `cd training/level_2 then cat checkpoint_beta.txt`

Completion: Submits a flag · flag is inside `/home/examiner/training/level_2/checkpoint_beta.txt`

Hint 1 (free): A sibling directory is reached by going UP first, then over. From your home directory the whole trip is one command: `cd training/level_2`.

Hint 2 (-5 XP): Once you are inside, run `cat checkpoint_beta.txt` and submit the find.

Instructor note: From inside `level_1` the sibling route is: `cd .. then cd level_2` — that is the lesson.

Warp Speed (Tab Completion)

15 pts · act1-tab · OK

Task: Typing full paths is slow, and a typo in a path is how the wrong file ends up in a report. Get yourself into your `Documents` directory without typing the whole name — make the shell finish it for you.

Answer: `cd Doc then cd Documents`

Completion: Auto-completes on: `allOf`

Hint 1 (free): Type `cd Doc`, press the Tab key — the shell completes it to `cd Documents` — then press Enter.

Also accepted: `cd ./Documents`, `cd ~/Documents`, `cd /home/examiner/Documents`

Instructor note: Students type `cd Doc` and press Tab. Typing it in full also counts.

Recall the Past (History)

10 pts · act1-history · OK

Task: Examiners repeat commands constantly, and retyping is how mistakes creep in. Bring back a command you have already run — the one that printed your position — and run it again without typing it out.

Answer: `pwd`

Completion: Auto-completes on: `allOf`

Hint 1 (free): Press the Up Arrow key to walk back through your command history until you reach `pwd`, then press Enter.

Instructor note: The lesson is the Up Arrow key, not the `pwd` command itself.

Act II: Reading the Evidence

Open the exhibits: file contents, true file types, and forensic checksums

Read the Case File

15 pts · act2-cat · OK

Task: Nothing else in this act makes sense until you have read the case file end to end. Put the whole of `Documents/case_1042.txt` on screen: the top says what Aurora Robotics reported, the bottom says what this lab expects of you.

Answer: `cat Documents/case_1042.txt`

Completion: Auto-completes on: allOf

Hint 1 (free): One command prints a whole file to the screen. It is called `cat`, short for concatenate.

Hint 2 (-5 XP): Run `cat Documents/case_1042.txt`.

Also accepted: `cat ./Documents/case_1042.txt`, `cat /home/examiner/Documents/case_1042.txt`

The Night Shift

15 pts · act2-head · OK

Task: `Documents/access.log` is the workstation timeline recovered from the seized laptop. Its first five entries alone show the machine being opened twice: once in daylight, and once at an hour when nobody was meant to be in the building. Put exactly those first five entries on screen.

Answer: `head -n 5 Documents/access.log`

Completion: Auto-completes on: outputEquals

Hint 1 (free): `head` prints the top of a file. One find says how many lines you want.

Hint 2 (-5 XP): Run `head -n 5 Documents/access.log`.

Also accepted: `head -5 Documents/access.log`, `head -n 5 ./Documents/access.log`

The Last Record

20 pts · act2-tail · OK

Task: A timeline is only as good as its ends. You have read the beginning; the final record is the one the intake sheet refers to, written when the laptop reached this lab. Read the end of `Documents/access.log` and submit the find on the last line.

Answer: `tail Documents/access.log`

Completion: Submits a flag · flag is inside `/home/examiner/Documents/access.log`

Hint 1 (free): `tail` prints the bottom of a file — the opposite end from `head`.

Hint 2 (-5 XP): Run `tail Documents/access.log`, then submit the find on the final line.

Don't Trust the Name

20 pts · act2-file · OK

Task: Exhibit `evidence/mystery_file` arrived with its extension stripped off. An examiner never identifies a file by its name — the answer is in the first few bytes of the file itself. Work out what this exhibit really is.

Answer: `file evidence/mystery_file`

Completion: Auto-completes on: `outputContains`

Hint 1 (free): One command reads a file's magic bytes and reports what it truly is.

Hint 2 (–5 XP): Run `file evidence/mystery_file`.

Also accepted: `file ./evidence/mystery_file`, `file /home/examiner/evidence/mystery_file`

Carving Text Out of a Binary

25 pts · act2-strings · OK

Task: `evidence/binary_data` is a compiled program, so `cat` gives you screenfuls of garbage. Pull the readable text out of it instead, and submit the recovery token hidden among the strings.

Answer: `strings evidence/binary_data`

Completion: Submits a flag · flag is inside `/home/examiner/evidence/binary_data`

Hint 1 (free): One command extracts printable character sequences from a binary.

Hint 2 (–5 XP): Run `strings evidence/binary_data`, then submit the recovery token in the output.

Integrity Checksum (MD5)

25 pts · act2-md5 · OK

Task: Evidence has to be hashed so the lab can prove nobody altered it — before examination and after. Compute the MD5 hash of `evidence/evidence.img`, then read the image header and submit the acquisition find recorded there.

Answer: `md5sum evidence/evidence.img then cat evidence/evidence.img`

Completion: Submits a flag · flag is inside `/home/examiner/evidence/evidence.img`

Hint 1 (free): One command computes an MD5 digest. Reading the header afterwards is an ordinary `cat`.

Hint 2 (–5 XP): Run `md5sum evidence/evidence.img`, then `cat evidence/evidence.img`.

Instructor note: The hash is NOT the flag — the flag is in the file text.

Act III: Following the Trail

Search, find, manual pages, and the WSL bridge to the seized laptop

Secrets in Plain Sight

25 pts · act3-grep · OK

Task: `Documents/secrets.txt` came off the seized laptop and only one line of it matters to this case. Find the line that carries `vault_passcode` and submit the find on it.

Answer: `grep vault_passcode Documents/secrets.txt`

Completion: Submits a flag · flag is inside `/home/examiner/Documents/secrets.txt`

Hint 1 (free): `grep` searches a file for a pattern. Give it the pattern first, then the file.

Hint 2 (-5 XP): Run `grep vault_passcode Documents/secrets.txt`.

Case-Insensitive Searching

30 pts · act3-grep · OK

Task: `Documents/logs.txt` records failures inconsistently: some lines say 'error', some 'ERROR'. A search that only matches one of them misses evidence. Find every error line with a single search, then submit the find one of them carries.

Answer: `grep -i "error" Documents/logs.txt`

Completion: Submits a flag · flag is inside `/home/examiner/Documents/logs.txt`

Hint 1 (free): One short `grep` find makes the match ignore letter case.

Hint 2 (-10 XP): Run `grep -i "error" Documents/logs.txt`.

Needle in /var/log

30 pts · act3-find · OK

Task: Aurora's badge readers write to this machine, somewhere under `/var/log`, and you do not know the filename. List every file under `/var/log` whose name ends in `.log`, then read the badge audit and submit the code recorded when badge 4417 was accepted at 02:14.

Answer: `find /var/log -name "*.log" then cat /var/log/badge_audit.log`

Completion: Submits a flag · flag is inside `/var/log/badge_audit.log`

Hint 1 (free): `find` walks a whole directory tree and can filter on the filename. Quote the wildcard pattern so the shell leaves it alone.

Hint 2 (-10 XP): Run `find /var/log -name "*.log"`, then `cat /var/log/badge_audit.log`.

The WSL Bridge

40 pts · act3-crossing · OK

Task: This bench runs Linux, but the seized laptop's Windows filesystem is mounted alongside it at `/mnt/c`. Read the evidence intake sheet at `/mnt/c/Users/Examiner/Desktop/CASE_FILES/intake.txt` and submit the verification find on it.

Answer: `cat /mnt/c/Users/Examiner/Desktop/CASE_FILES/intake.txt`

Completion: Submits a flag · flag is inside `/mnt/c/Users/Examiner/Desktop/CASE_FILES/intake.txt`

Hint 1 (free): WSL mounts each Windows drive under `/mnt`, so drive C: shows up as `/mnt/c`.

Hint 2 (-10 XP): Run `cat /mnt/c/Users/Examiner/Desktop/CASE_FILES/intake.txt`.

The Bridge, Unassisted

45 pts · act3-crossing-solo · OK

Task: The examiner before you left handover notes on the Windows side, at

`C:\Users\Examiner\Documents\handover_notes.txt`. Reach them from this Linux shell and submit the bridge token inside. Nobody is going to translate the path for you this time.

Answer: `cat /mnt/c/Users/Examiner/Documents/handover_notes.txt`

Completion: Submits a flag · flag is inside `/mnt/c/Users/Examiner/Documents/handover_notes.txt`

Hint 1 (free): Translate the Windows path yourself: `C:\` becomes `/mnt/c/`, and every backslash becomes a forward slash.

Hint 2 (-15 XP): Run `cat /mnt/c/Users/Examiner/Documents/handover_notes.txt`.

Instructor note: Deliberately unassisted: the student must translate the Windows path themselves.

Consult the Manual

30 pts · act3-man · OK

Task: You are about to need `evtrace` to replay everything the exhibits recorded, and left alone it replays one source. Nobody has told you the option that changes that, and a manual page is where a tool says what it can do. This bench also logs every tool it runs against an exhibit by the reference printed in that tool's manual.

Answer: `man evtrace`

Completion: Submits a flag

Hint 1 (free): Every real Linux tool ships a manual page, and one command opens it. Read the OPTIONS section while you are in there, because the next challenge needs what is in it.

Hint 2 (-10 XP): Run `man evtrace`. The bench reference is in DESCRIPTION, and the option that replays every recorded source is in OPTIONS.

Instructor note: Flag is inside the DESCRIPTION section.

Install the Bench Tool (apt-get)

35 pts · act3-apt · OK

Task: `evtrace` is not installed on this workstation yet. Install it from the package repository, then replay every recorded source and submit the code the badge reader stream prints.

Answer: `sudo apt-get update then sudo apt-get install evtrace -y then evtrace -a`

Completion: Submits a flag · flag is inside `evtrace`

Hint 1 (free): Refresh the package lists first, then install the package. Both steps need administrator rights, and each one is its own command line. The manual told you which option replays every source.

Hint 2 (-10 XP): Run `sudo apt-get update`, then `sudo apt-get install evtrace -y`, then `evtrace -a`.

Act IV: The Pipeline

Pipes, redirection, filters, and multi-stage analysis

Filter the Noise (Inverted Grep)

40 pts · act4-grep-v · OK

Task: Documents/network_stream.log is almost all ordinary allowed traffic, and the one line that matters is buried in it. Show only the packets that were NOT allowed, then submit the find on the CRITICAL_DATA_LEAK entry.

Answer: `grep -v "ALLOW" Documents/network_stream.log`

Completion: Submits a flag · flag is inside /home/examiner/Documents/network_stream.log

Hint 1 (free): `grep` normally keeps the lines that match. One find inverts that and keeps everything else.

Hint 2 (-10 XP): Run `grep -v "ALLOW" Documents/network_stream.log`.

First Pipeline (grep | wc -l)

45 pts · act4-pipe-count · OK

Task: Your report needs a number, not a listing: how many packets in Documents/network_stream.log were not allowed? Get that count onto the screen and nothing else. A pipeline is the classic way to do it, but this challenge grades the number you produced, so any command that prints the right count scores.

Answer: `wc -l then grep -v "ALLOW" Documents/network_stream.log | wc -l then grep -vc ALLOW Documents/network_stream.log`

Completion: Auto-completes on: outputEquals

Hint 1 (free): The `|` operator feeds one command's output straight into the next, and `wc -l` counts lines.

Hint 2 (-10 XP): Run `grep -v "ALLOW" Documents/network_stream.log | wc -l`. The shorter `grep -vc ALLOW Documents/network_stream.log` reaches the same number and is accepted too.

Also accepted: `grep -v "ALLOW" Documents/network_stream.log | wc -l`, `grep -vc ALLOW Documents/network_stream.log`, `grep -vc "ALLOW" Documents/network_stream.log`, `cat Documents/network_stream.log | grep -v ALLOW | wc -l`

Extracting Fields (cut and grep)

45 pts · act4-pipe-csv · OK

Task: Documents/security_events.csv is comma separated, and the rows tagged FLAG_EMIT carry a token in column 6. Isolate those rows, cut out just that column, and submit the token.

Answer: `grep "FLAG_EMIT" Documents/security_events.csv | cut -d, -f6`

Completion: Submits a flag · flag is inside /home/examiner/Documents/security_events.csv

Hint 1 (free): Two commands joined by a pipe: `grep` picks the rows, `cut` picks the column. `cut` needs to be told the delimiter and the field number.

Hint 2 (-10 XP): Run `grep "FLAG_EMIT" Documents/security_events.csv | cut -d, -f6`.

Output Redirection (>)

50 pts · act4-redirect · OK

Task: Findings that only exist on screen are not findings. Pull every error line out of `Documents/logs.txt` — however it is capitalised — and save them to a new file at `/tmp/errors.log` instead of printing them. Then read that file back to confirm the capture worked.

Answer: `grep -i "error" Documents/logs.txt > /tmp/errors.log` then `cat /tmp/errors.log`

Completion: Auto-completes on: allOf

Hint 1 (free): `>` sends output into a file instead of the screen. Do not forget the option that makes the search ignore case, or you will miss lines.

Hint 2 (-10 XP): Run `grep -i "error" Documents/logs.txt > /tmp/errors.log`, then `cat /tmp/errors.log`.

Also accepted: `grep -i "error" Documents/logs.txt > /tmp/errors.log`, `cat Documents/logs.txt | grep -i error > /tmp/errors.log`

Instructor note: The redirect prints nothing — that is correct.

Act V: Closing the Case

Read the partition table, carry the offset, carve the container

Read the Partition Table

60 pts · act5-scan · OK

Task: `evidence/seized_drive.raw` is the raw image of the drive that left the building, and an encrypted container is somewhere inside it. `mmls` is the Sleuth Kit tool that prints a disk image's partition table: where each partition starts, how long it is, and what kind it is. Nothing can be carved until you know which sector the container begins at.

Answer: `mmls evidence/seized_drive.raw`

Completion: Auto-completes on: allOf

Hint 1 (free): `mmls` takes the path to the image and nothing else. Then read the Start column of the row that says Encrypted Container.

Hint 2 (-10 XP): Run `mmls evidence/seized_drive.raw`. The container is slot 002, and the Start it reports is the number the next challenge wants.

Also accepted: `mmls ./evidence/seized_drive.raw`, `mmls /home/examiner/evidence/seized_drive.raw`

Instructor note: Students must READ the Start sector offset from the table; the capstone no longer prints it for them.

Carve the Container (Finish Line)

120 pts · act5-capstone · OK

Task: This is the whole case in one command line. `dd` copies a REGION of a file rather than all of it: `bs=512` makes each of its blocks one sector, and `skip=` counts how many of those blocks to step over before it starts reading. Hand it the sector `mmls` gave you and it lifts the container out of `evidence/seized_drive.raw`; hand it a different number and it copies the wrong bytes without complaining.

Answer: `dd if=evidence/seized_drive.raw of=container.img bs=512 skip=41 then cat container.img`

Completion: Auto-completes on: allOf

Hint 1 (free): `dd` reads with `if=`, writes with `of=`, and needs two more things: how big a block is, and how many blocks to step over before it starts. A sector is 512 bytes.

Hint 2 (-20 XP): Run `dd if=evidence/seized_drive.raw of=container.img bs=512 skip=41`, then `cat container.img` to read what came out.

Also accepted: `dd if=evidence/seized_drive.raw of=container.img bs=512 skip=41 count=4`, `dd bs=512 skip=41 if=evidence/seized_drive.raw of=container.img`, `dd if=/home/examiner/evidence/seized_drive.raw of=/home/examiner/container.img bs=512 skip=41`

Instructor note: Chains off act5-scan. A wrong offset is rejected by the tool, so a student who skipped the scan must go back.

Act VI: The Seized Laptop (Windows)

Windows CMD parity: `dir`, `/a`, `type`, `findstr`, `attrib`, and `certutil`

Windows Navigation (`dir` and `cd`)

15 pts · act6-nav · OK

Task: You are now working the seized laptop from its own Windows command line. Start the same way you started on Linux: show what is in `C:\Users\Examiner`.

Answer: `dir`

Completion: Auto-completes on: allOf

Hint 1 (free): In `cmd.exe` the listing command is `dir`. Typed on its own, `cd` prints where you are instead of moving you.

Hint 2 (-5 XP): Run `dir`.

Also accepted: `dir .`

Hidden File Attributes (`attrib`)

20 pts · act6-attrib · OK

Task: Windows hides a file with an attribute bit, not with a leading dot, so the Linux trick you learned in Act I will not find it here. Inspect the attributes of `evidence\mystery_file`, then read it and recover the find.

Answer: `attrib evidence\mystery_file` then `type evidence\mystery_file`

Completion: Submits a flag · flag is inside `C:\Users\Examiner\evidence\mystery_file`

Hint 1 (free): `attrib` shows a file's attribute letters, and H means Hidden. `type` prints a file the way `cat` does on Linux.

Hint 2 (-5 XP): Run `attrib evidence\mystery_file`, then `type evidence\mystery_file`.

Windows String Search (`findstr /i`)

25 pts · act6-findstr · OK

Task: The laptop's event log extract is at `Documents\logs.txt`. Find the evidence marker in it, ignoring letter case, and submit the find. There is no `grep` on Windows.

Answer: `findstr /i "marker" Documents\logs.txt`

Completion: Submits a flag · flag is inside `C:\Users\Examiner\Documents\logs.txt`

Hint 1 (free): The `cmd.exe` search command is `findstr`. Its switches start with a forward slash, and `/i` ignores case.

Hint 2 (-5 XP): Run `findstr /i "marker" Documents\logs.txt`.

CertUtil Hashing

20 pts · act6-certutil · OK

Task: Finish the chain of custody on the Windows side: compute the MD5 hash of `evidence\evidence.img`. Windows ships no `md5sum`, so use the tool that is already on every Windows machine.

Answer: `certutil -hashfile evidence\evidence.img MD5`

Completion: Auto-completes on: allOf

Hint 1 (free): `certutil` needs the `-hashfile` switch, then the file, then the name of the algorithm.

Hint 2 (-5 XP): Run `certutil -hashfile evidence\evidence.img MD5`.

Also accepted: `certutil -hashfile evidence/evidence.img MD5`, `certutil -hashfile evidence\evidence.img md5`

Accepted command variants

Auto-completing challenges accept the natural variations a beginner types, so a correct command is never silently ignored:

- Optional quotes around a path
- Path prefixes: `./Documents` , `~/Documents` , or the full absolute path
- Short paths when already inside the folder
- Flag spacing: `head -n 5` , `head -n5` , `head -5`
- Windows: forward or back slashes, any capitalization (`md5` or `MD5`)
- Trailing spaces are ignored everywhere

Teaching notes

- **The Coach** (toggle in the terminal title bar) explains every command and error in plain language. Explanations fade after the second use of a command. Leave it ON for beginners.
- **Reference tab** holds a manual page for every command, and tells students the real-shell equivalent. Point stuck students there before they buy a hint.
- **Unsimulated tools** reply with what they are, rather than a bare "command not found".
- **Hints:** the first is free and conceptual; the one that gives the exact command costs XP. Hint use is visible in the instructor dashboard as a signal of where the class is struggling.
- **Watch the stuck points.** The dashboard flags any challenge with a solve rate under 35%. Now that briefs no longer contain the answer, that number means something: it marks a concept the class did not get, not a typing error.